

EXECUTIVE SUMMARY

KDL Ú PRO VJP 123

**Spending peaks in December and stress concentrates in two of five segments;
six supportive tools can reach 89.4% of customers**

Task 1 Spending patterns

December is the peak, driven by more purchases, not bigger ones.

It holds 15.1% of annual spend vs 5.3% in February: transactions +187.6%, average ticket -2.1%. Stressed months put 71.0% of spending into discretionary items vs 39.0% in healthy months.

Task 2 Financial health

Stress is rare and short, usually set off by one big purchase.

Only 0.9% of months score below 40, and 95.7% recover the next month. 73.4% of those months contain one purchase $\geq$ 20% of monthly income, vs 0.9% of months scoring 60–80.

Task 3 Engagement

"High" engagement is the norm, so it barely separates customers.

99.0% of customer-months are "high" or "very high", yet POS still carries 58.8% of transactions. 59 full-year customers (6.5%) are among the healthiest but least engaged.

Task 4 Segmentation

Five segments; two carry almost all financial stress.

Stable & Engaged 32.6% · Stretched & Engaged 20.8% · Lower Engagement & Pressured 19.8% · Emerging Digital 17.6% · Limited History 9.1%. The two pressured segments hold 65 of the 70 customers who ever scored below 40.

Task 5 Recommendations

Six supportive tools reach 893 customers (89.4%).

Start with spend alerts (389 customers; strongest link to stress months), then planning reminders before December (687 customers, 68.8%).

Guardrail: Financial_health_score is used only to decide who gets support first. It is never used to deny credit, cut a credit limit, or block an account.

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INTRODUCTION TO THE CASE

Business context

Who:

ITB, a consumer finance company in Vietnam, offering cards and short-term credit to retail customers.

What ITB wants to know:

- How customers spend and how they use each channel
- How financially healthy they are
- How these patterns differ across customer groups, and how health and engagement relate
- Whether distinct customer segments exist
- What practical actions can support engagement and financial wellbeing

Our job:

Answer these five questions with the data, without using `financial_health_score` for any credit decision.

Dataset overview

Transactions 2025

1,852,394 rows

One row per transaction:
category, merchant, channel,
payment method, amount, time.

Consumer-month 2025

10,992 rows

One row per customer per month:
financial health score, engagement
score, spending ratios, synthetic
income, credit and balance.

999 customers, linked by `consumer_id` across calendar year 2025: 908 have all 12 months, 91 have only 1–2 months. They live in 34 provinces and hold 396 job titles.

How we read the data

- **Synthetic amounts:** all data is synthetic, so we compare customers with ratios and percentiles, not VND amounts.
- **Health bands:** below 40 = stressed · 40–60 · 60–80 · 80+ = healthy
- **Big purchase:** one transaction worth 20% or more of the customer's own monthly income
- **Digital channels:** every channel except POS (QR, e-commerce, mobile app, recurring payment). **Online** follows the data's online flag, which excludes QR. Each share states whether it is counted by transactions or by spend.
- **Two bases:** customer-level findings use the 908 full-year customers; the 91 with only 1–2 months are kept as a separate group. Channel counts and target sizes cover all 999.

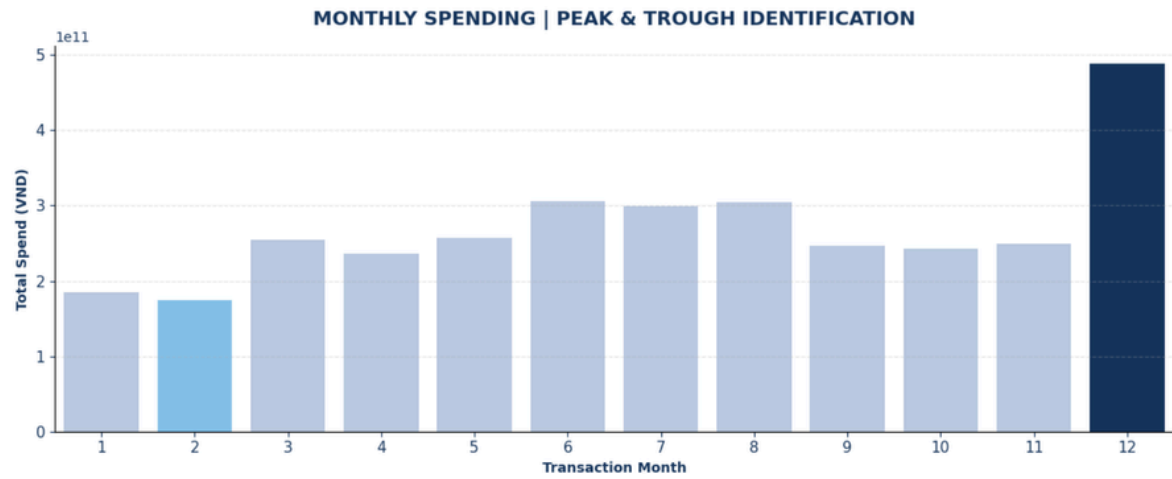
Guardrail: `Financial_health_score` is used only to decide who gets support first. It is never used to deny credit, cut a credit limit, or block an account.

MACRO TEMPORAL ANALYSIS

Consumption growth is driven by transaction volume rather than larger purchases

EXPLORATORY DATA ANALYSIS

PEAK MONTH & TROUGH MONTH



PEAK MONTH & TROUGH MONTH

Annual Total Spend: 3,244,632,958,000 VND

PEAK MONTH	TROUGH MONTH
Month 12	Month 2
Total Spend: 488,032,709,000 VND	Total Spend: 174,365,111,000 VND
Annual Share: 15.04%	Annual Share: 5.37%

December records the highest spend at VND 488.03B (15.04% of annual spend), versus February at VND 174.37B (5.37%), the lowest.

December therefore captures nearly 3× February’s share of annual spending, indicating pronounced seasonal concentration.

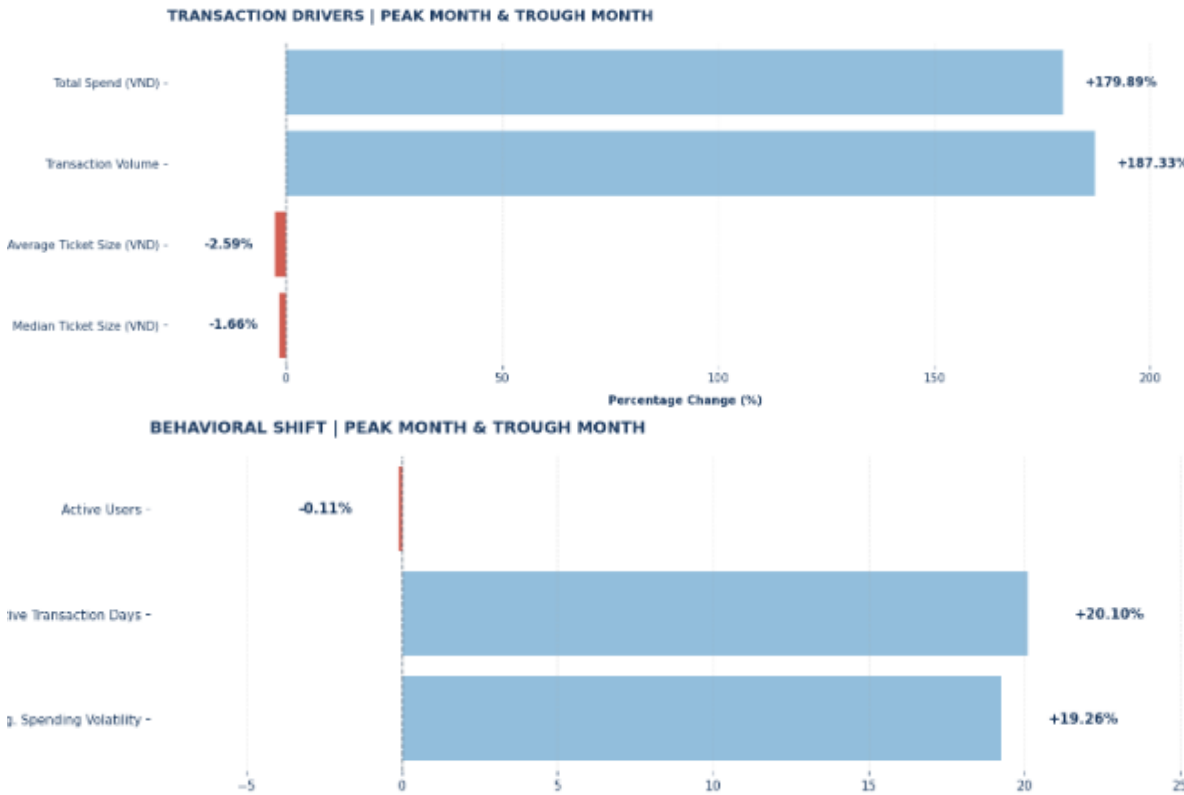
TRANSACTION DRIVERS & BEHAVIOR SHIFT

TRANSACTION DRIVERS | PEAK MONTH (M12) vs. TROUGH MONTH (M2)

	Delta (%)	12	2
Total Spend (VND)	179.89%	488,032,709,000.00	174,365,111,000.00
Transaction Volume	187.33%	280,598.00	97,657.00
Average Ticket Size (VND)	-2.59%	1,739,259.40	1,785,485.02
Median Ticket Size (VND)	-1.66%	1,182,000.00	1,202,000.00

BEHAVIORAL SHIFT | PEAK MONTH vs TROUGH MONTH

	Delta (%)	12	2
Active Users	-0.11%	918.00	919.00
Avg. Active Transaction Days	20.10%	30.39	25.30
Avg. Spending Volatility	19.26%	1.73	1.45



DECOMPOSITION OF SPENDING DRIVERS: PEAK MONTH BEHAVIORAL SHIFTS

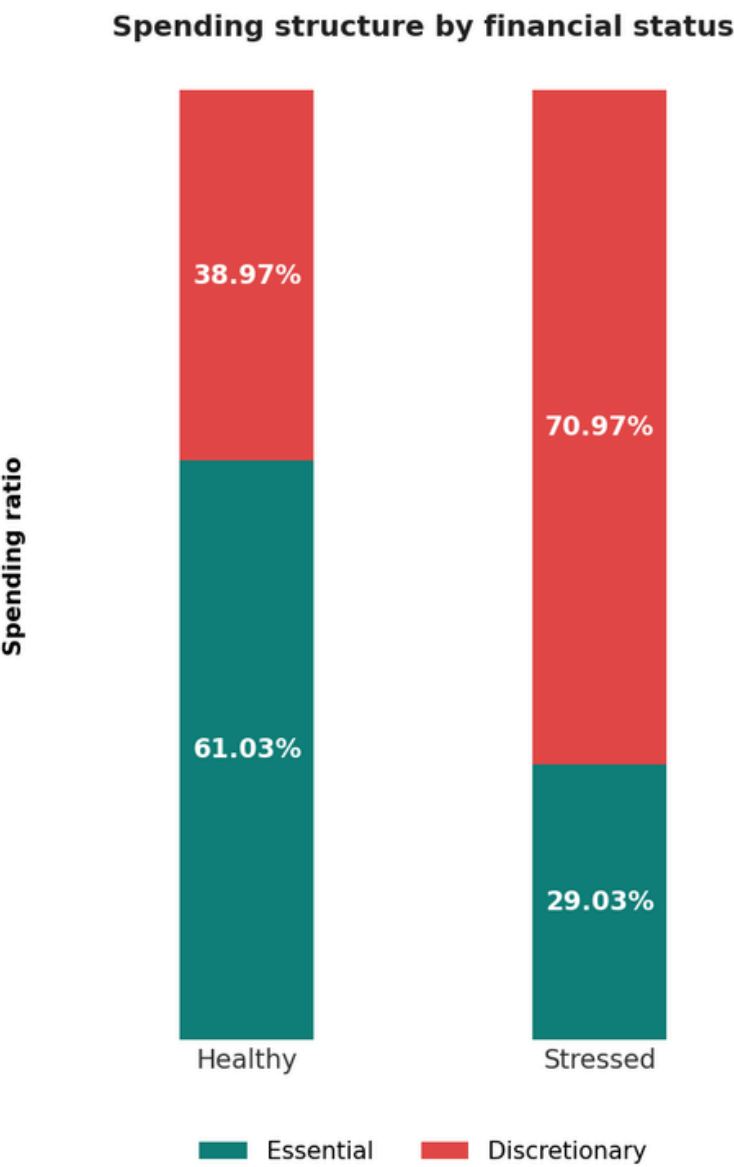
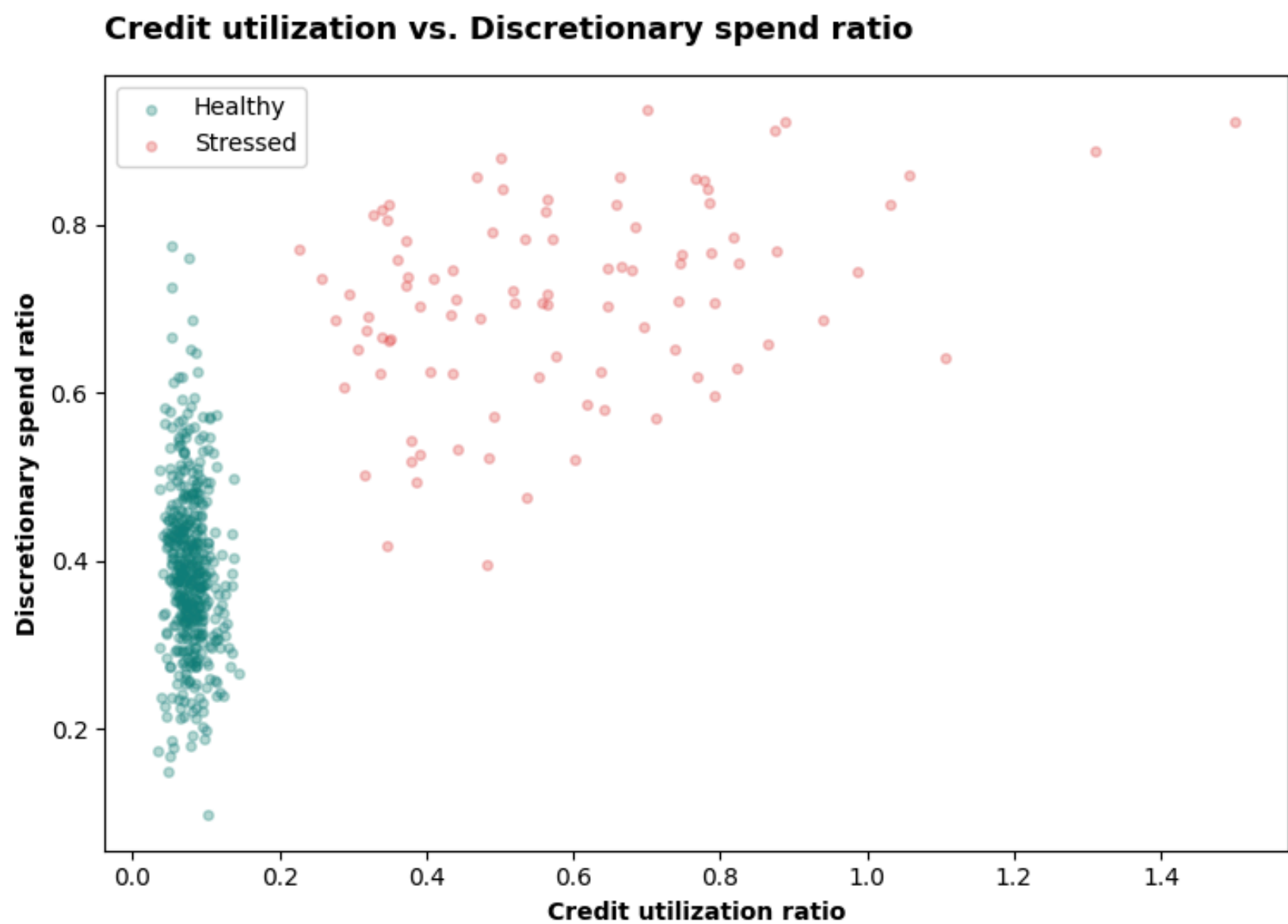
Strategic Assertion (Business Insight)	Quantitative Evidence & Behavioral Translation
Structural Shift to High-Frequency Micro-Purchasing	Total Spend surged by 179.89% purely as a function of exploding Transaction Volume (+187.33%), completely diverging from the marginal contractions in both Average (−2.59%) and Median (−1.66%) Ticket Sizes. Behaviorally, this structural shift indicates the revenue uplift is driven by high-frequency, fragmented purchasing rather than premium or bulk buying.
Growth via Deepened Cohort Engagement	Active Users remained essentially flat (−0.11%), while Avg. Active Transaction Days rose 20.10%. The peak month reflects intensified engagement from the existing user base rather than market expansion, as retained customers transacted significantly more often.
Elevated Budget Volatility & Erratic Spending	Despite the massive revenue surge, Avg. Spending Volatility increased by 19.26%. This reveals that peak-month consumption is characterized by concentrated, highly erratic spending bursts rather than evenly distributed, sustainable daily expenditures.

source: task1_1

STRESSED CUSTOMERS SPEND 71% ON DISCRETIONARY NEEDS

An opportunity for guided budgeting, not credit restriction

Financially healthy customers allocate 61% of spend to essentials; financially stressed customers flip this ratio, allocating 71% to discretionary spend. This signals an unmet need for budgeting tools and financial guidance, framing an engagement opportunity rather than a credit risk to restrict.



71 %

of Stressed customers' spend goes to discretionary needs

61 %

of Health customers' spend goes to esstential needs

THE DIGITAL ADOPTION GAP: WHY TOP ECONOMIC HUBS LAG IN DIGITAL CONVERSION

Despite generating leading transaction volumes, Ho Chi Minh city and Ha Noi severely lag behind the global digital benchmark. This gap is not driven by digital illiteracy, but by a structural reliance on legacy POS systems in daily retail categories.

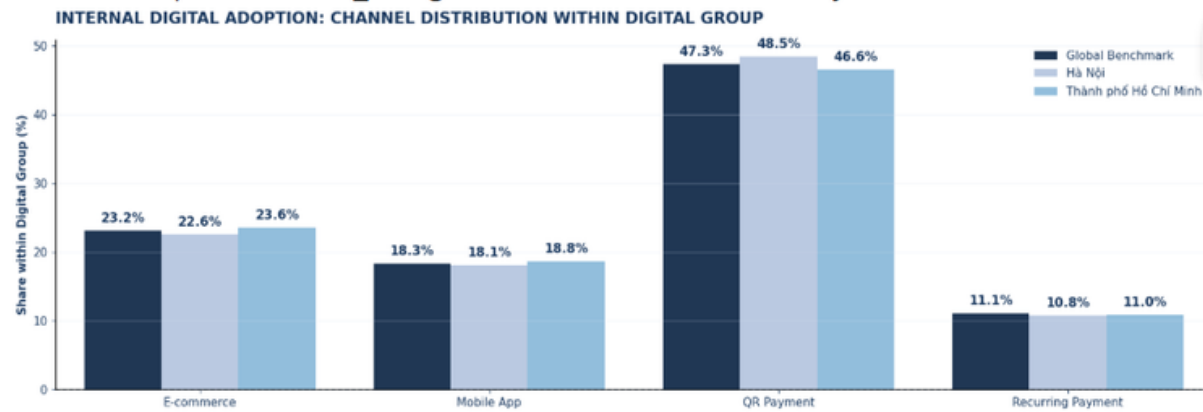
Spatial Identification & The Adoption Gap

SPATIAL PROVINCES: HIGH-VOLUME LAGGING HUBS

* Benchmark Metrics:
- High Volume Threshold (Median Total Spend): 82,693,483,500 VND
- Global Digital Ratio: 41.46%

Tỉnh / Thành phố (Province / City)	Total Spend (VND)	Digital Ratio (%)
Thành phố Hồ Chí Minh	442,175,772,000	41.30%
Hà Nội	267,264,508,000	40.91%
Đồng Nai	183,072,471,000	41.35%
Lâm Đồng	135,389,980,000	41.30%
Cần Thơ	126,027,485,000	41.15%
Hưng Yên	116,442,487,000	41.34%
Vĩnh Long	106,026,604,000	41.32%
Đắk Lắk	90,836,537,000	41.31%

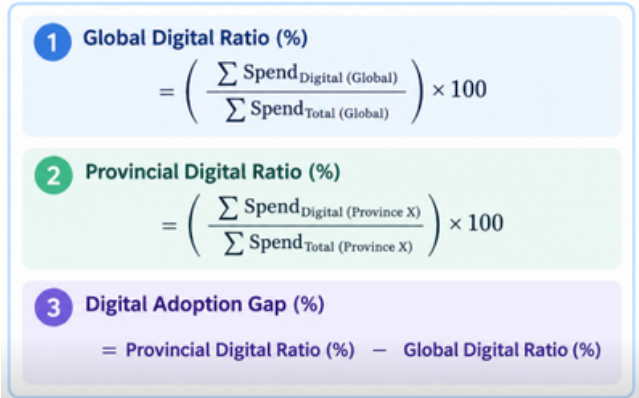
Category Breakdown & Supply-Side Bottlenecks



The digital lag is a merchant-side infrastructure bottleneck trapped within physical retail, not a lack of consumer digital literacy

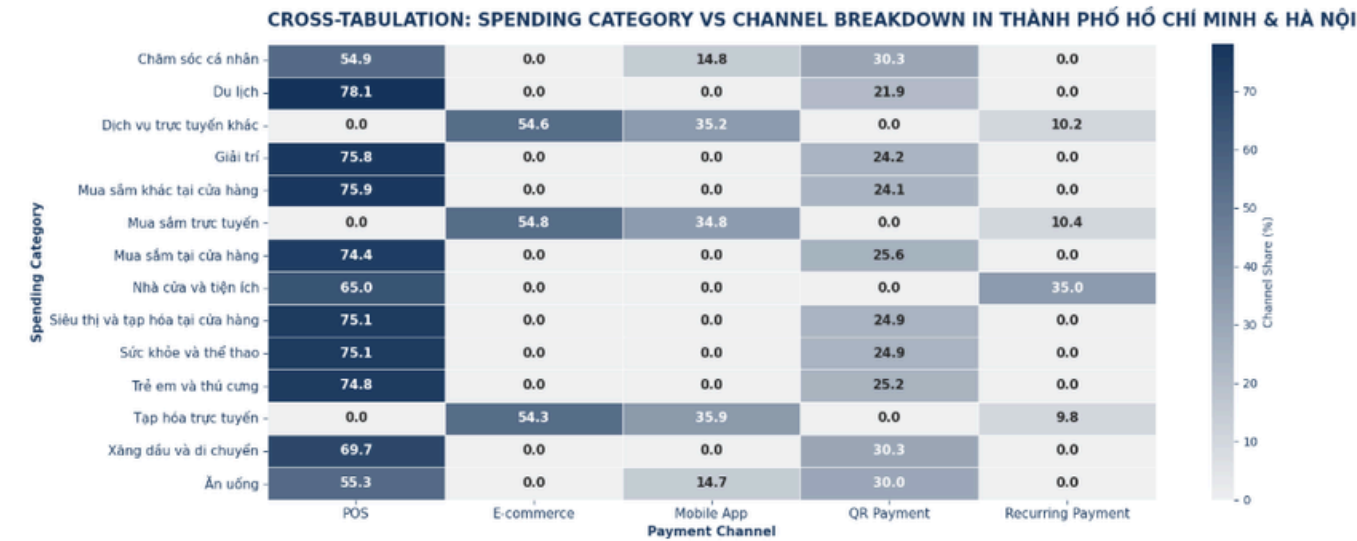
Cross-Level Validation & Demographic Friction

Overall engagement scores in both cities (~77.8) perfectly match the global benchmark, confirming a highly active user base. However, Hanoi’s noticeably older demographic (Median age 52 vs global 48), concentrated in specialized/traditional occupations, adds a compounding layer of generational friction to its digital conversion rate.



Ho Chi Minh City and Hanoi operate as high-volume economic engines burdened by a structural digital adoption deficit

espite generating a massive combined volume of over 709 Billion VND, both hubs trail the 41.46% global digital benchmark (Hanoi at 40.91%, HCMC at 41.30%). This digital gap is mathematically offset by their inflated reliance on traditional POS infrastructure (~59% channel share)



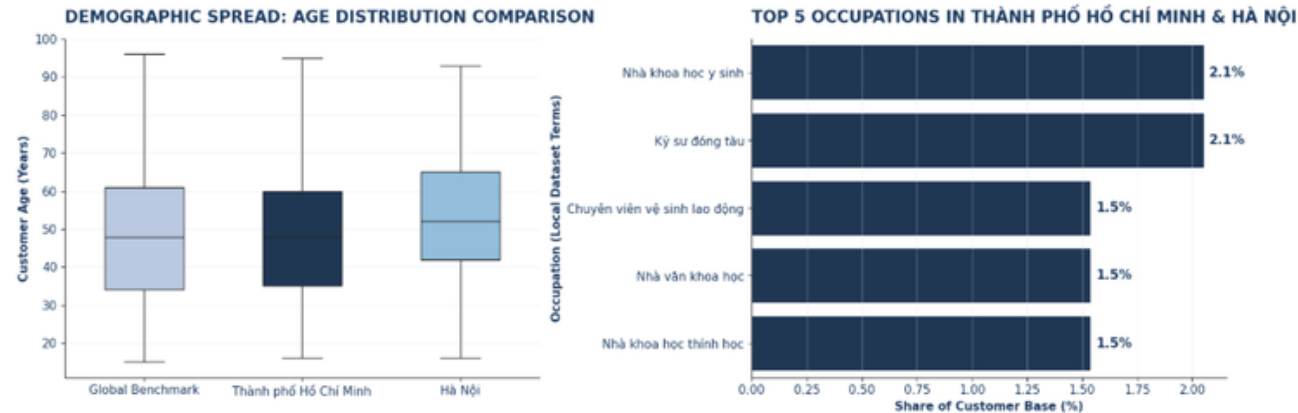
EXPLORATORY DATA ANALYSIS

CHANNEL DECOMPOSITION & DIGITAL ADOPTION GAP

Metrics / Channels	Global Benchmark	Thành phố Hồ Chí Minh	Hà Nội
Total Spend (VND)	3,244,632,958,000	442,175,772,000	267,264,508,000
Digital Ratio (%)	41.46%	41.30%	40.91%
Digital Adoption Gap	0.00%	-0.17%	-0.55%
POS Share	58.54%	58.70%	59.09%
E-commerce Share	9.62%	9.76%	9.23%
Mobile App Share	7.61%	7.74%	7.40%
QR Payment Share	19.63%	19.26%	19.85%
Recurring Payment Share	4.61%	4.53%	4.43%

Coss-tabulation reveals that POS dominance is heavily anchored by in-store physical categories (Supermarkets and In-store Shopping command >75% POS share).

Meanwhile, internal digital distribution perfectly mirrors the global baseline (QR Payments lead at ~47%), proving customers readily adopt digital channels wherever local merchant ecosystems permit.



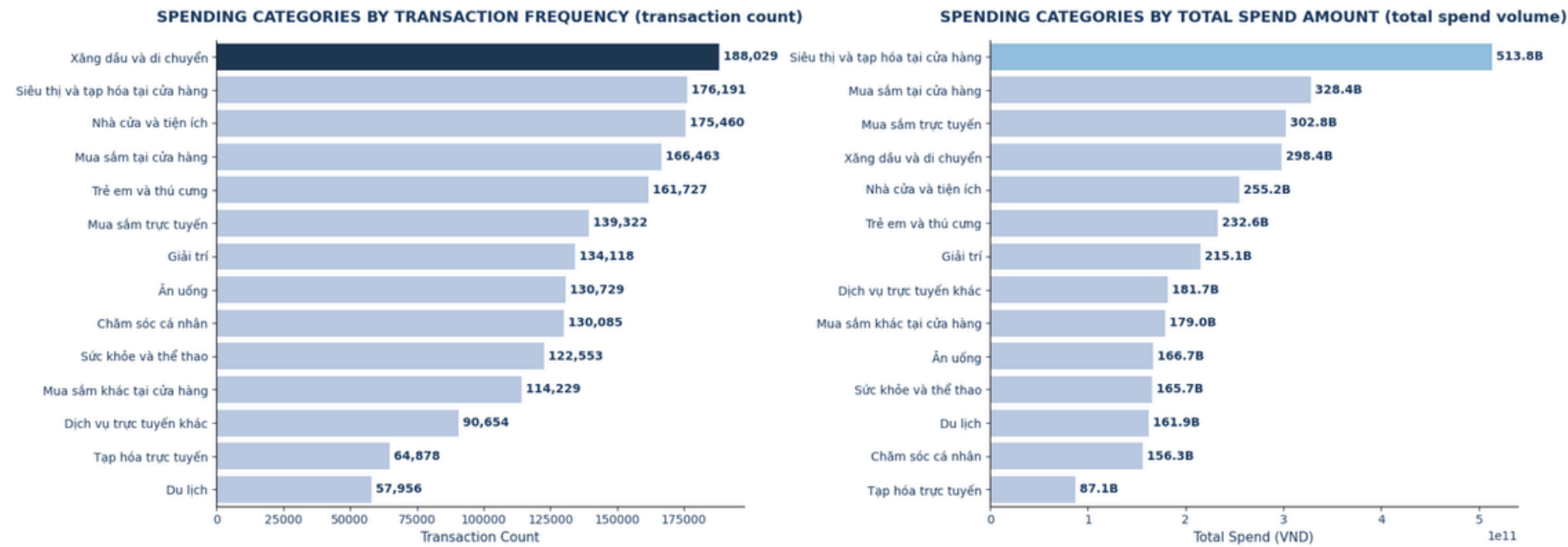
The adoption gap is predominantly an environmental constraint rather than user disengagement, though Hanoi faces an additional generational hurdle

UNCOVERING THE HABIT-REVENUE DUALITY IN CONSUMER SPENDING

Fuel & Transport drives daily habit loops, while Grocery serves as the core revenue anchor.

EXPLORATORY DATA ANALYSIS

Strategic Duality: Habit Hook vs Revenue Anchor



Category Polarization

- Fuel & Transport acts as the primary engagement hook (188,029 transactions)
- Supermarket & Grocery serves as the core revenue anchor (513.8B VND in total spend volume)

Consumer Universality: Base-Wide Behavioral Habit

CROSS-LEVEL BEHAVIORAL CONCENTRATION AUDIT

Customer Transaction Category	Penetration (%)	Avg. Active Days / Month	Avg. Spending Category Diversity
Xăng dầu và di chuyển (Customer Base)	98.67%	28.77	13.71
Siêu thị và tạp hóa tại cửa hàng (Customer Base)	99.68%	28.67	13.67

Cross-level validation confirms that both categories achieve near-absolute market penetration (98.67% and 99.68%) and identically high engagement rhythm (~28.7 active days/month).

This proves they are daily foundational habits for the entire customer base, demonstrating how the same users intuitively switch between high-frequency micro-transactions (Transport) and high-value planned shopping (Supermarket).

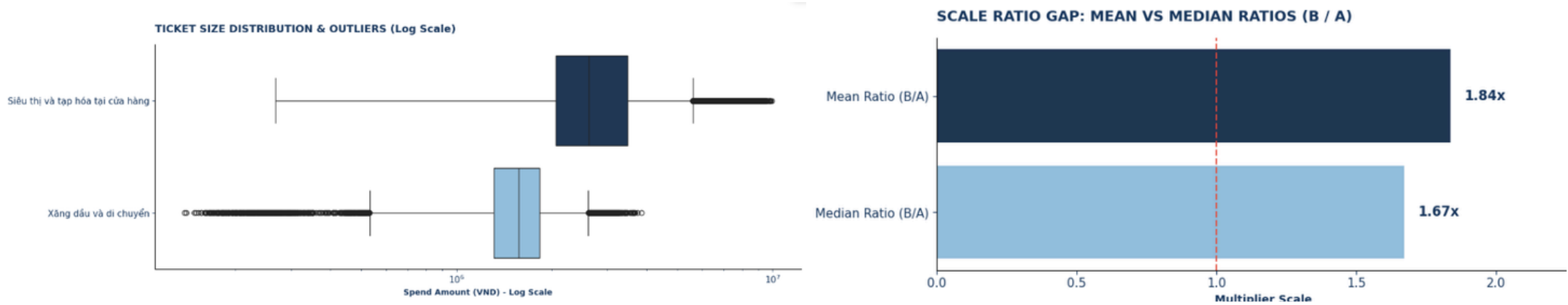
The behavioral divergence between these categories stems from universal consumer mental accounting, not from isolated or distinct customer segments.

Basket Dynamics: High-Value Outliers vs. Micro-Stability

TICKET SIZE DISTRIBUTION ANALYSIS: CATEGORY A VS CATEGORY B

Spending Category	Mean Ticket Size (VND)	Median Ticket Size (Q2 - VND)	Interquartile Range (IQR - VND)	Skewness Coeff. (Skew)
Siêu thị và tạp hóa tại cửa hàng	2,916,003	2,626,000	1,412,000	1.0946
Xăng dầu và di chuyển	1,586,933	1,572,000	521,000	0.2673

Supermarket's heavy right-skewness (1.09 Skew) and wide dispersion (1.41M VND IQR) confirm the presence of extreme high-value outlier purchases, whereas Transport remains rigidly symmetric and stable (0.26 Skew, 521K VND IQR)



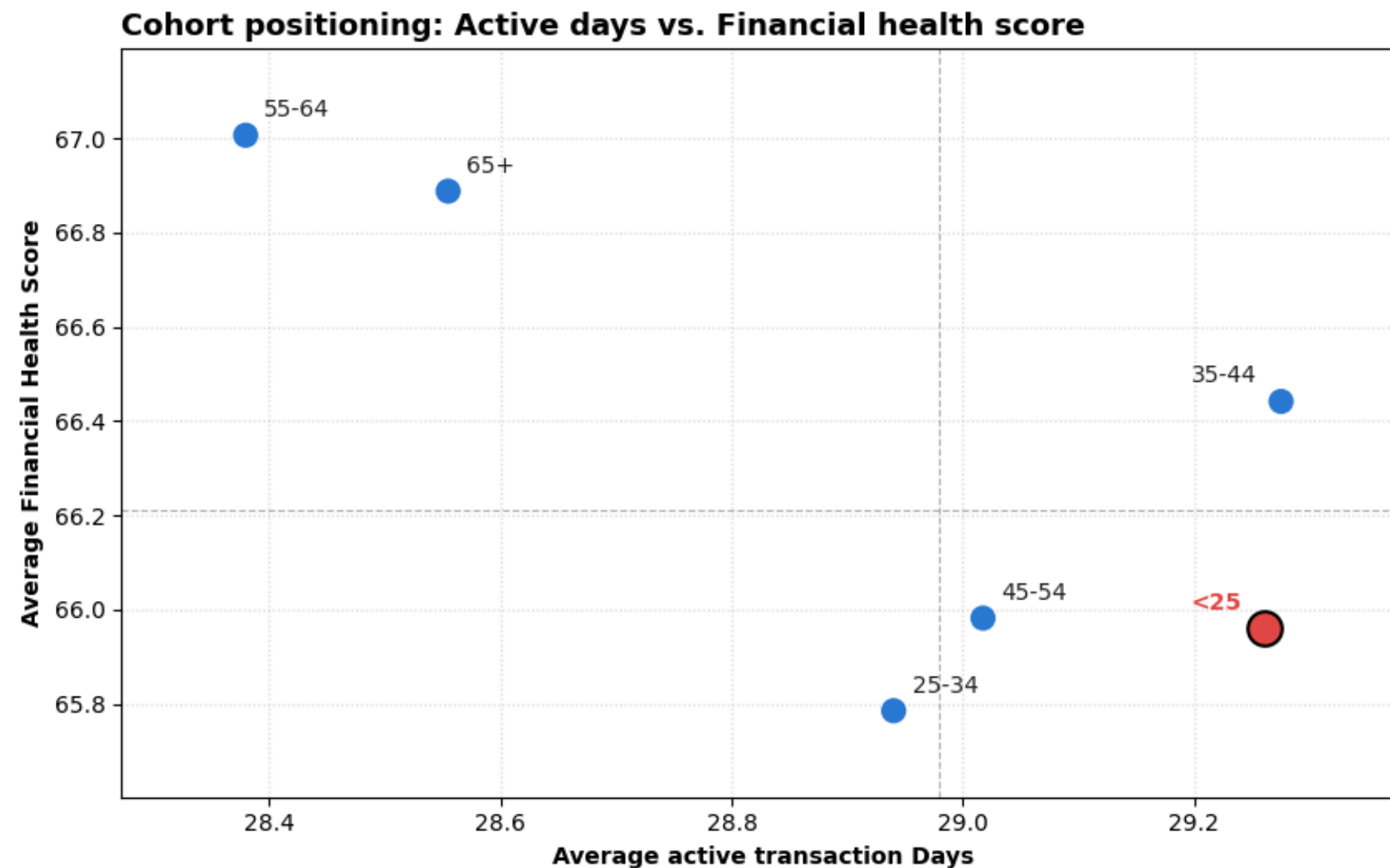
Supermarket spending demands significantly higher liquidity, exhibiting a 1.67x Median ticket size premium (2.62M vs 1.57M VND). Transport remains rigidly symmetric and stable (0.26 Skew, 521K VND IQR)

Supermarket revenue is driven by high-value, variable bulk purchasing, sharply contrasting with the highly predictable, routine micro-spending of Transport

AGE ALONE DOESN'T EXPLAIN FINANCIAL HEALTH

Under-25 customers are highly active, but engagement should drive the segmentation

The under-25 cohort combines above-median transaction activity with the lowest health score among active cohorts — likely reflecting early-career financial constraints rather than poor money habits. However, ANOVA confirms age explains only 0.25% of health-score variance (statistically significant only due to large sample size) — age alone is too weak a lever for segmentation.



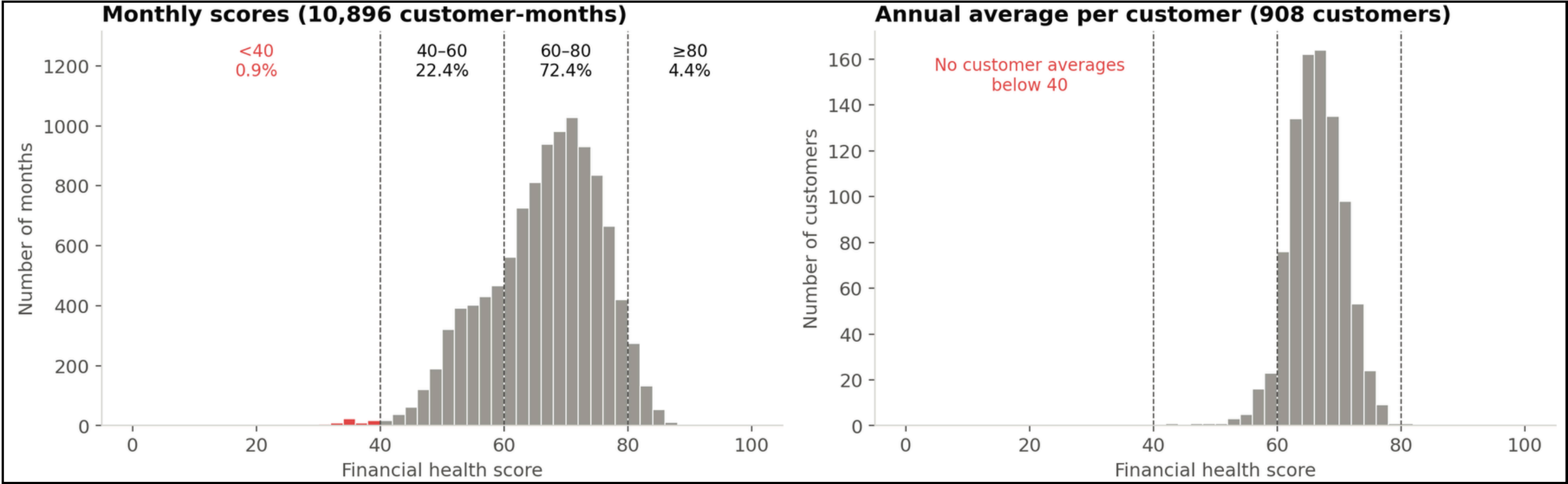
< 25

Most active cohort with the lowest health score among active segments

0.25 %

Of health score variance explained by age ($p < 0.001$, significant only due to sample size)

FINANCIAL STRESS IS RARE AND SHORT-LIVED: UNDER 1% OF MONTHS FALL BELOW 40, AND 95.7% RECOVER THE NEXT MONTH



0.9% of 10,896 customer-months scored below 40

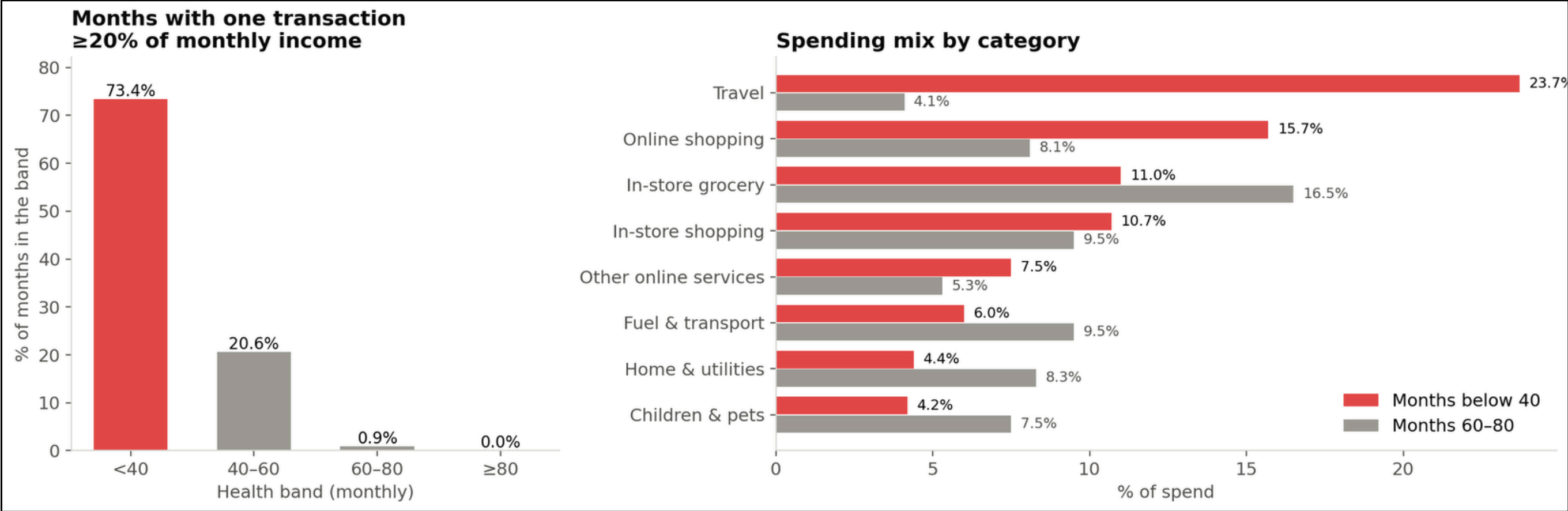
69 customers had at least one month below 40

95.7% of months below 40 are followed by a month at 40 or above

0 customers average score below 40 over the year (lowest: 42.3)

Definition	Scores are grouped into the brief's four bands (<40, 40–60, 60–80, ≥80) in two ways: by month (10,896 customer-months) and by customer, using each customer's 12-month average. Recovery means the next month scores 40 or above; The 95.7% excludes December, which has no next month.
Source	consumer_financial_health_engagement_2025 · Base: 908 customers observed all 12 months · Details: Task 2 notebook, Part 2 and Part 3

LOW-HEALTH MONTHS USUALLY CONTAIN ONE LARGE PURCHASE, MOSTLY TRAVEL



87.8%

of score variation explained by 4 monthly ratios

65.6%

of large-purchase money in low-health months goes to travel

46 of 94

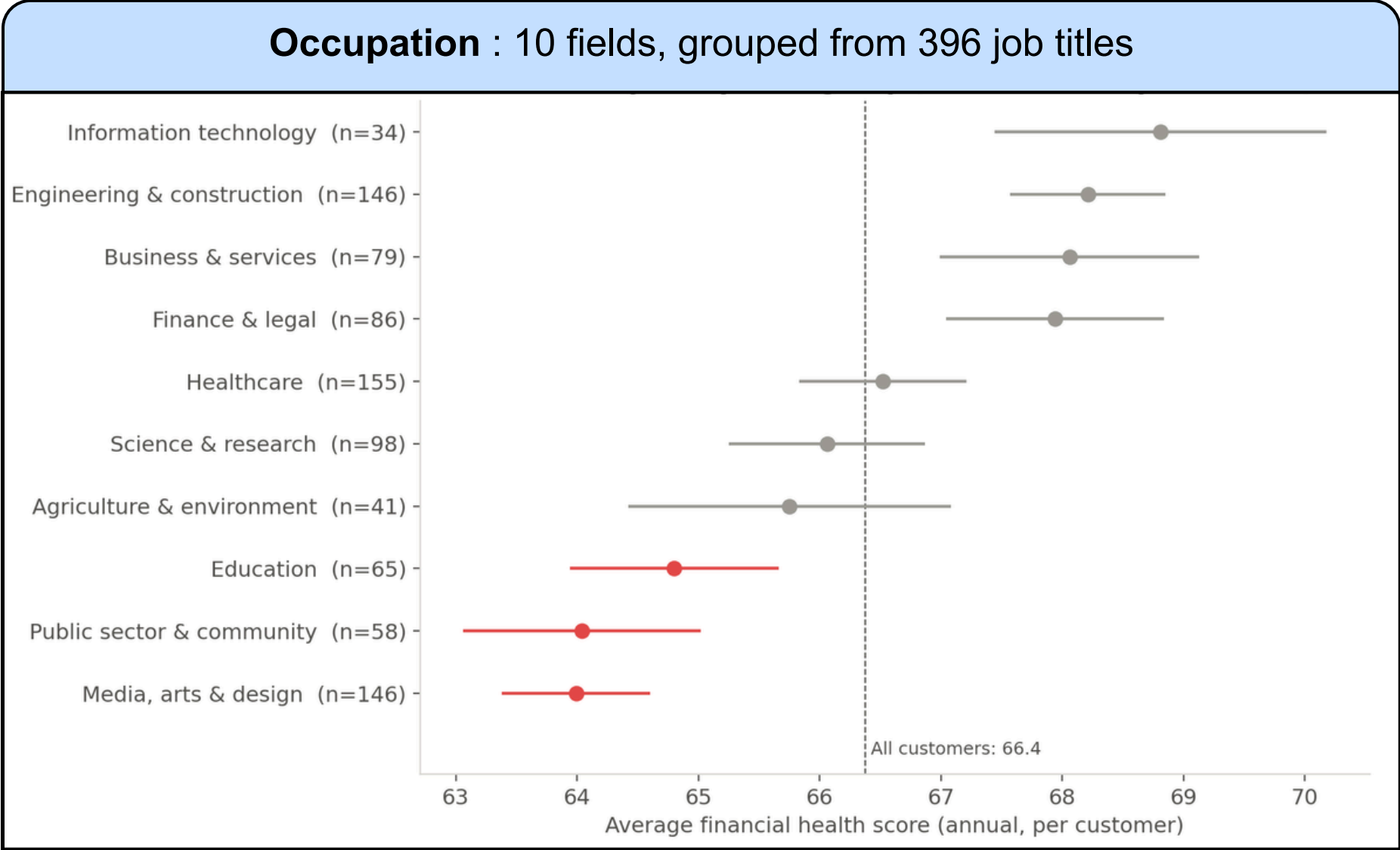
low-health months fall in December or January

39.7%

of spending in low-health months happens 18:00–24:00

Definition	A large purchase = one transaction worth ≥ 20% of that customer's own monthly income; using 10% or 30% instead gives the same conclusion. 87.8% = R ² of a linear regression of the monthly score on spend-to-income, credit utilization, spending volatility and essential share.
Source	consumer_financial_health_engagement_2025, consumer_transactions_2025 • Base: 908 full-year customers • Details: Task 2 notebook, Part 1 and Part 3

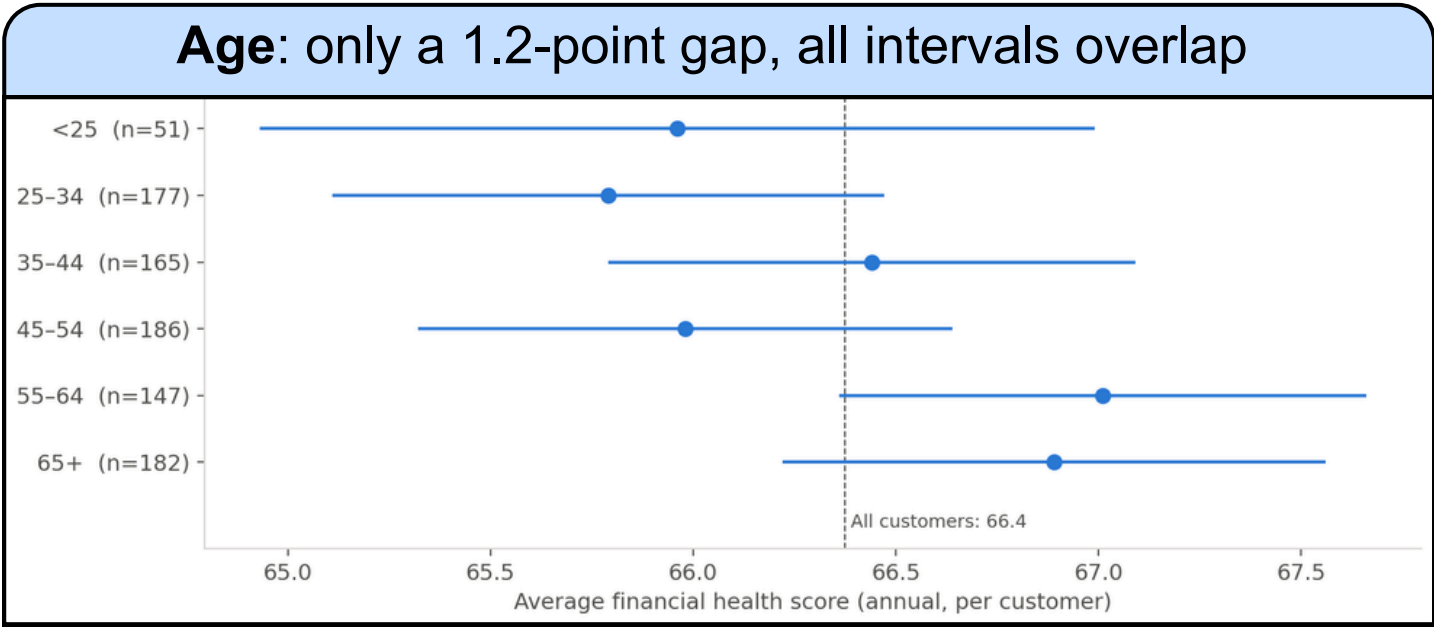
FINANCIAL HEALTH DIFFERS MOST BY OCCUPATION, LESS BY PROVINCE AND BARELY BY AGE



4.8 points gap between the highest field (IT, 68.8) and the lowest (Media & arts, 64.0).

49.3% of customers with a month below 40 come from the three lowest-scoring fields, which hold only 29.6% of all customers.

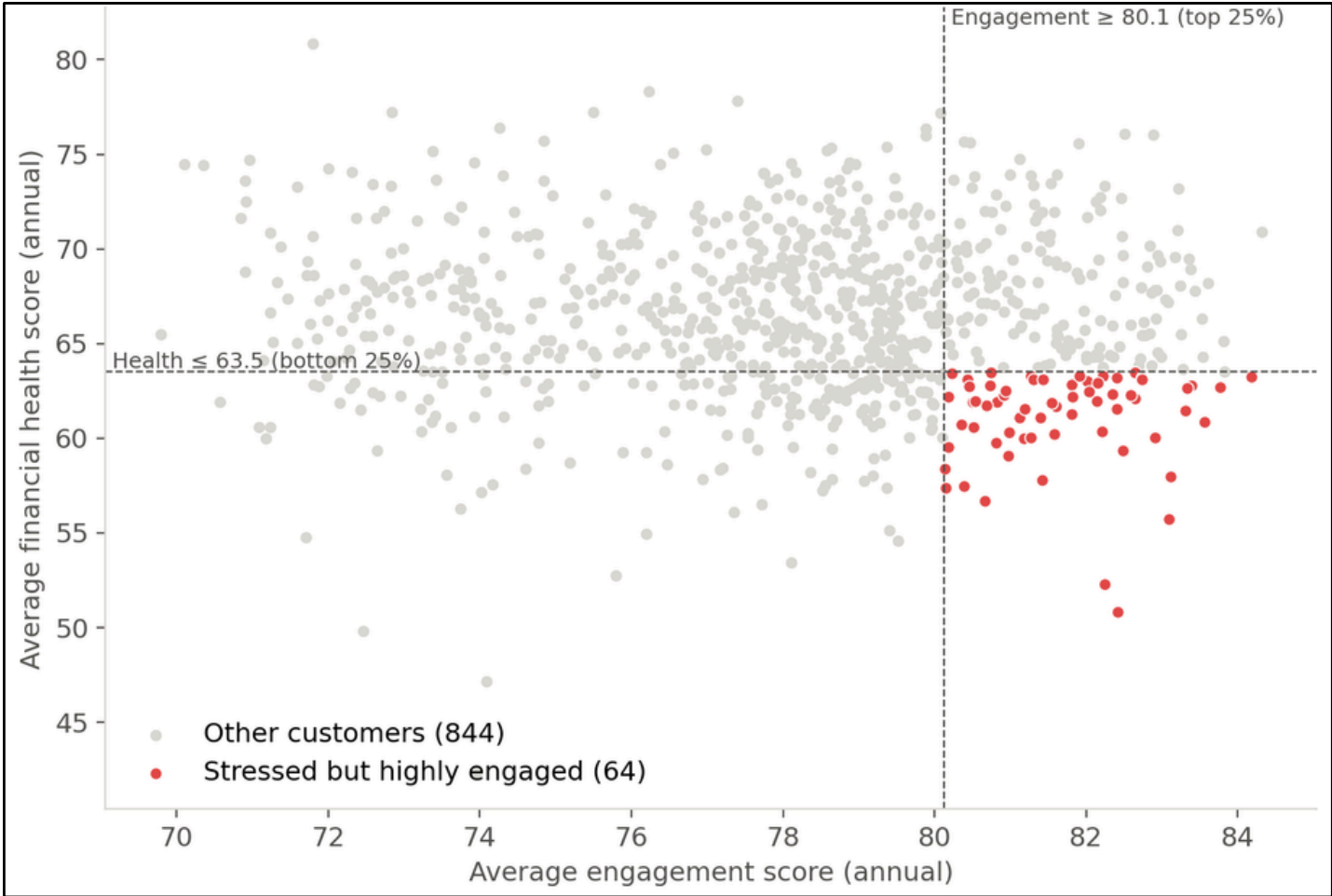
Note Their income sits at the 44th percentile vs 52nd for other fields; while spending is the same (50th).



Province	3.9-point gap, driven by the two largest cities. Hà Nội (68.1) & HCMC (68.0) score highest; Đà Nẵng is lowest (64.2). Most provinces in between overlap
Definition	Dots = average of each customer's 12-month score. Gap = highest group average minus lowest; lines show 95% confidence intervals. All three splits pass the Kruskal-Wallis test (p < 0.05). Percentile = share of the 908 customers with lower income (or spending). Provinces are compared only where there are 20+ customers (19 of 34). Province data is synthetic.
Source	consumer_financial_health_engagement_2025 • Base: 908 full-year customers • Details: Task 2 notebook, Part 4

64 HIGHLY ENGAGED CUSTOMERS ARE UNDER STEADY PRESSURE BECAUSE THEY SPEND MORE, NOT BECAUSE THEY EARN LESS

Rule	Bottom 25% on financial health (≤ 63.5) and top 25% on engagement (≥ 80.1), using annual averages of 908 full-year customers $\rightarrow$ 64 customers (7.0%) Why relative cut-offs: no customer averages below 40, and 99% of months are already "high" or "very high" engagement
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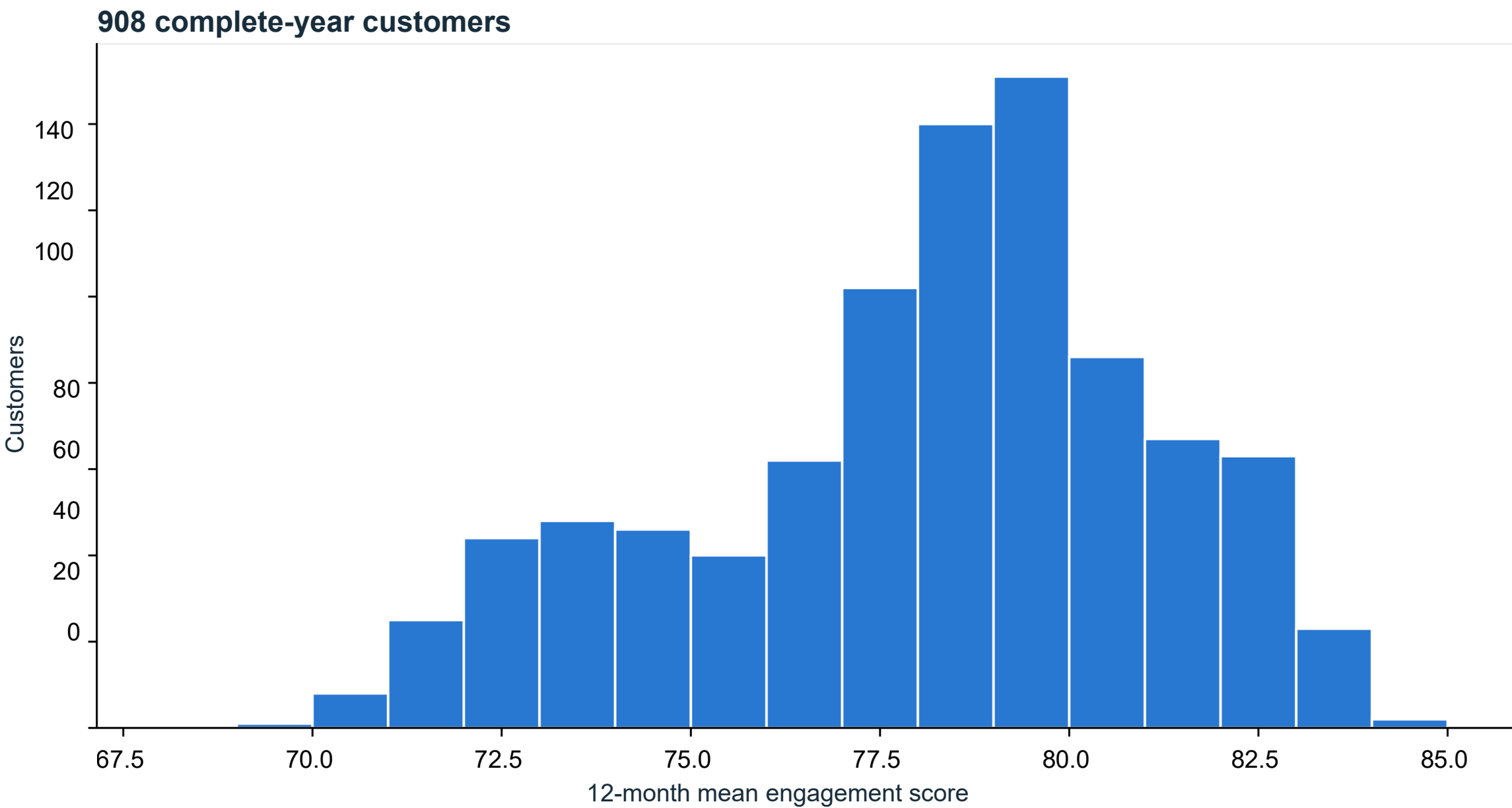
Why 25%? Other cut-offs tested		
Cut-off	Customer	Verdict
10%	8 (0.9%)	Too few to profile
25%	64 (7%)	Chosen
33%	114 (12.6%)	Health cut-off 64.4, close to the 66.4 average

Typical customer (median): group vs others		
Measure	Group of 64	Other 844
Age	37.5	49
Transaction per month	244	182
Spending / income	0.83	0.69
Months spending above income (per year)	3	1
Spending percentile	79th	48th
Income percentile	62nd	49th

Overspending in more months than others, not one big trip	They spend more than they earn in 3 months of the year, while travel is only 1.2% of their spending (others: 2.2%) so the pressure is not from trips.
Media & arts and education again, but with higher income	42.2% of this group work in these two fields (23.3% of all customers). On slide 11 these fields scored low because they earn less; this group spends more.
Definition	"Stressed" here means in the bottom 25% on health score, not <40 : only 7 of the 64 ever had a month below 40. Profile figures are medians.

Engagement scores are saturated - behavioral depth is needed to distinguish customers

99.0% of observed customer-months score ≥ 60 ; distinguish customer shares from monthly observations.



908 / 999

customers observed in all 12 months

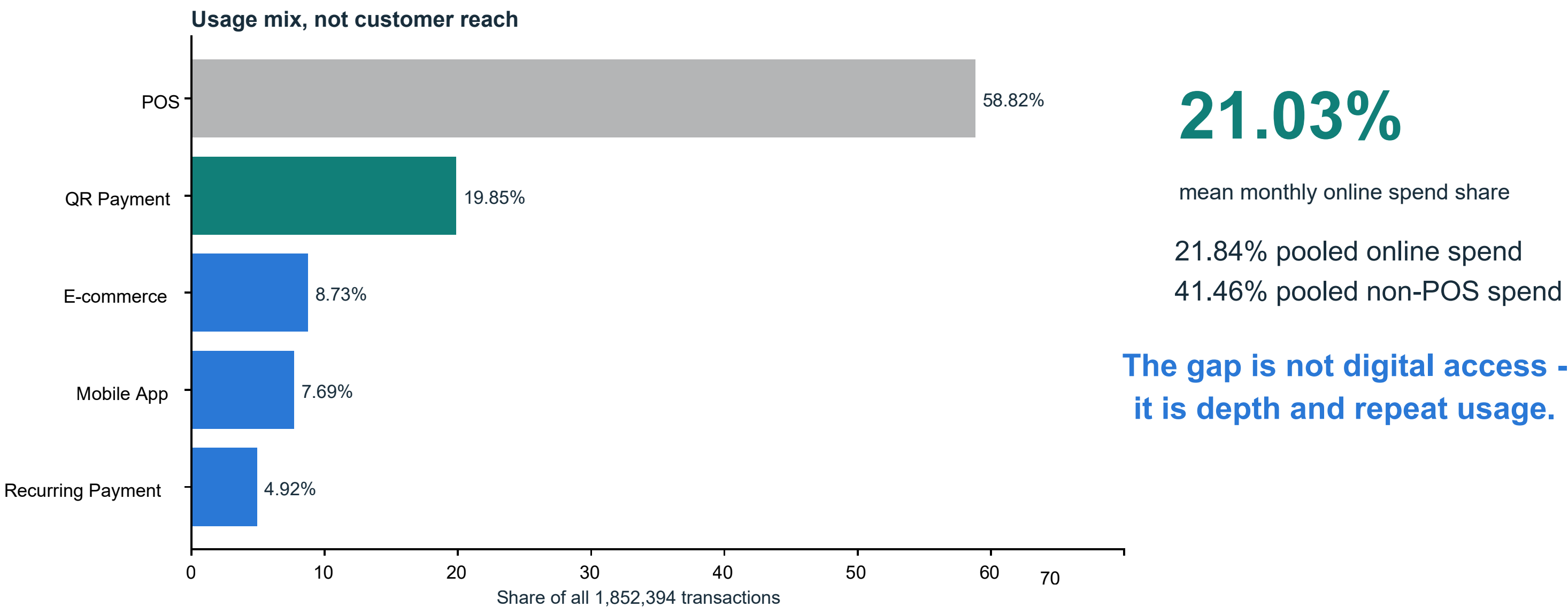
Low <40 : 0 (0.0%)
Moderate 40- <60 : 0 (0.0%) High
60- <80 : 667 (73.5%) Very high
 ≥ 80 : 241 (26.5%)

High engagement is the norm,
not a differentiator.

The 91 short-history customers are retained in all-month analysis,
not treated as comparable full-year customers or automatically labeled as churned.

Digital reach is nearly universal, yet POS still dominates actual usage.

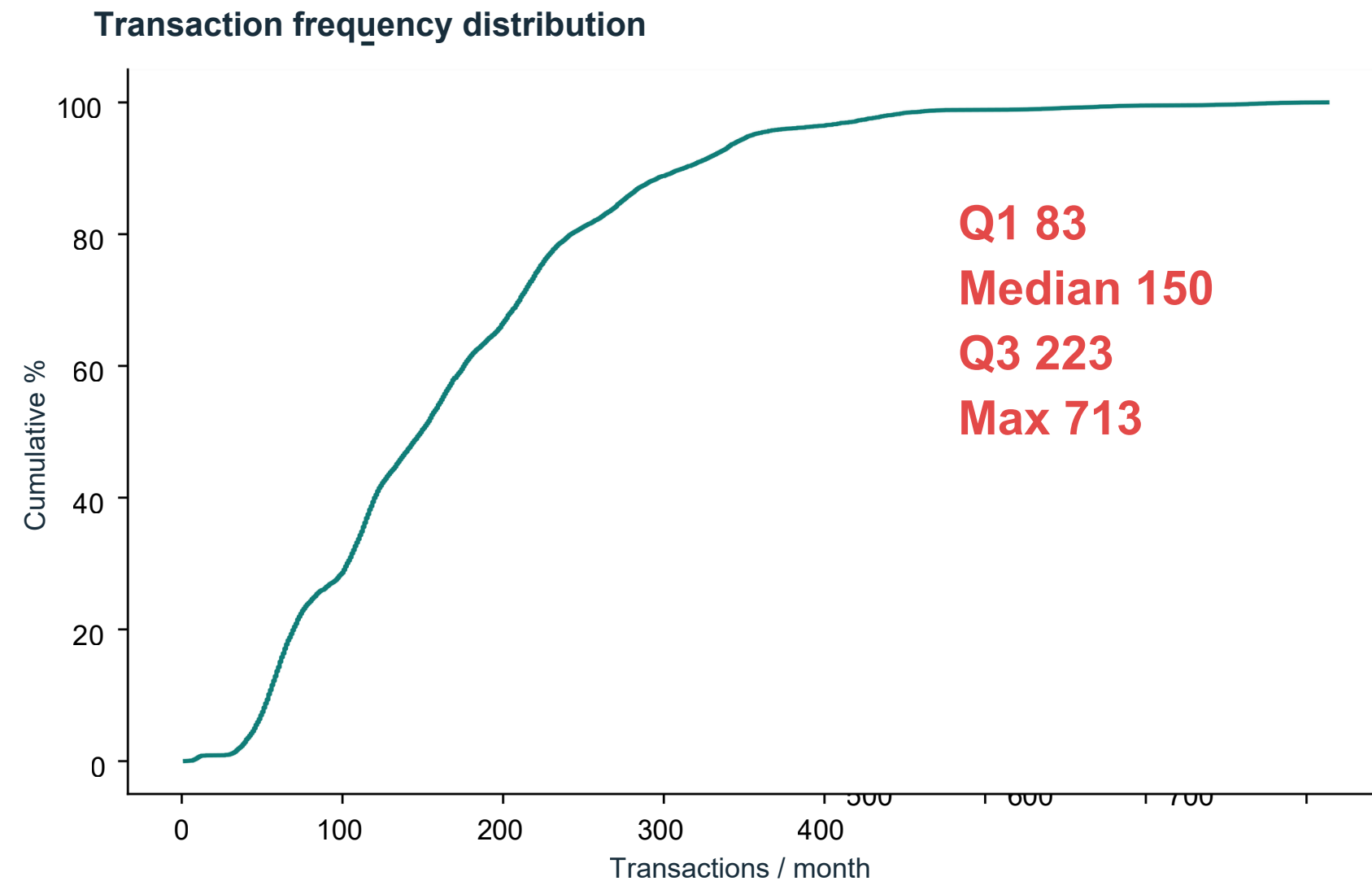
Annual reach = at least one use / 999 customers. Online excludes QR; broad digital includes all non-POS channels.



Annual customer reach: POS: 100.0% | QR: 97.6% | E-commerce: 99.0% | Mobile App: 98.1% | Recurring: 94.3%

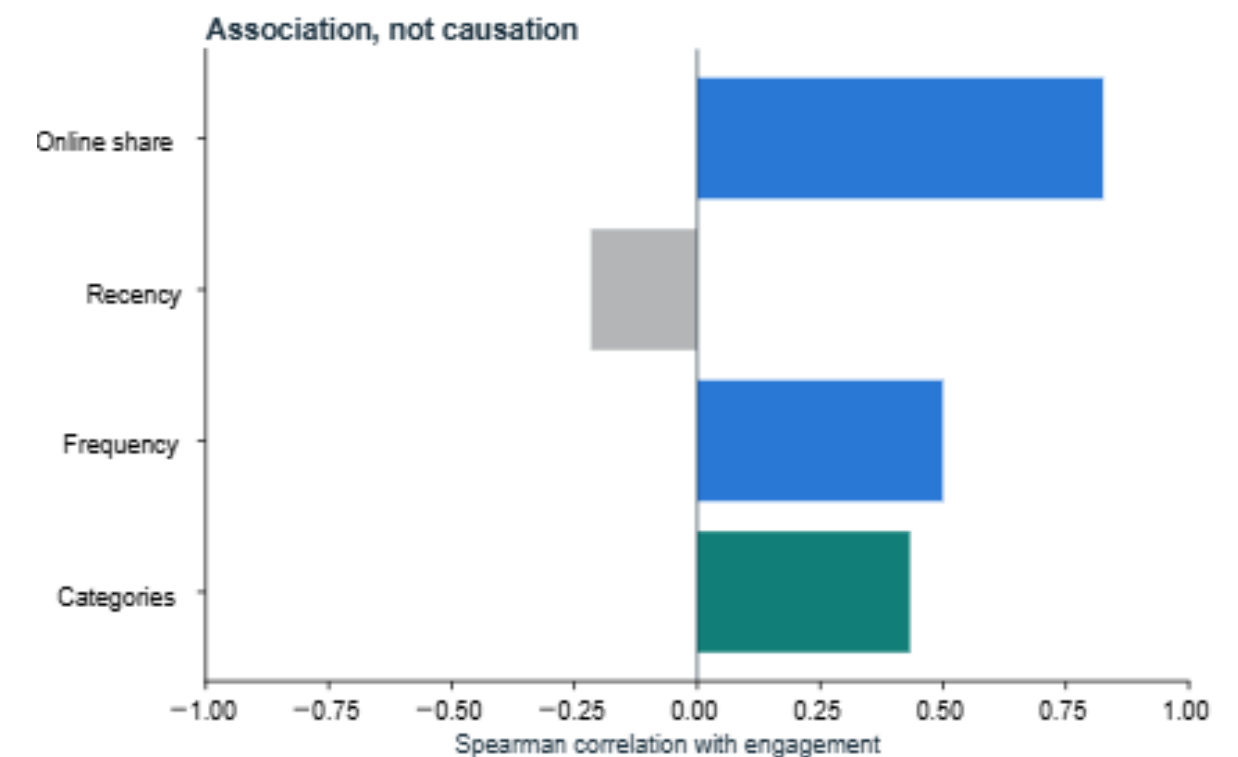
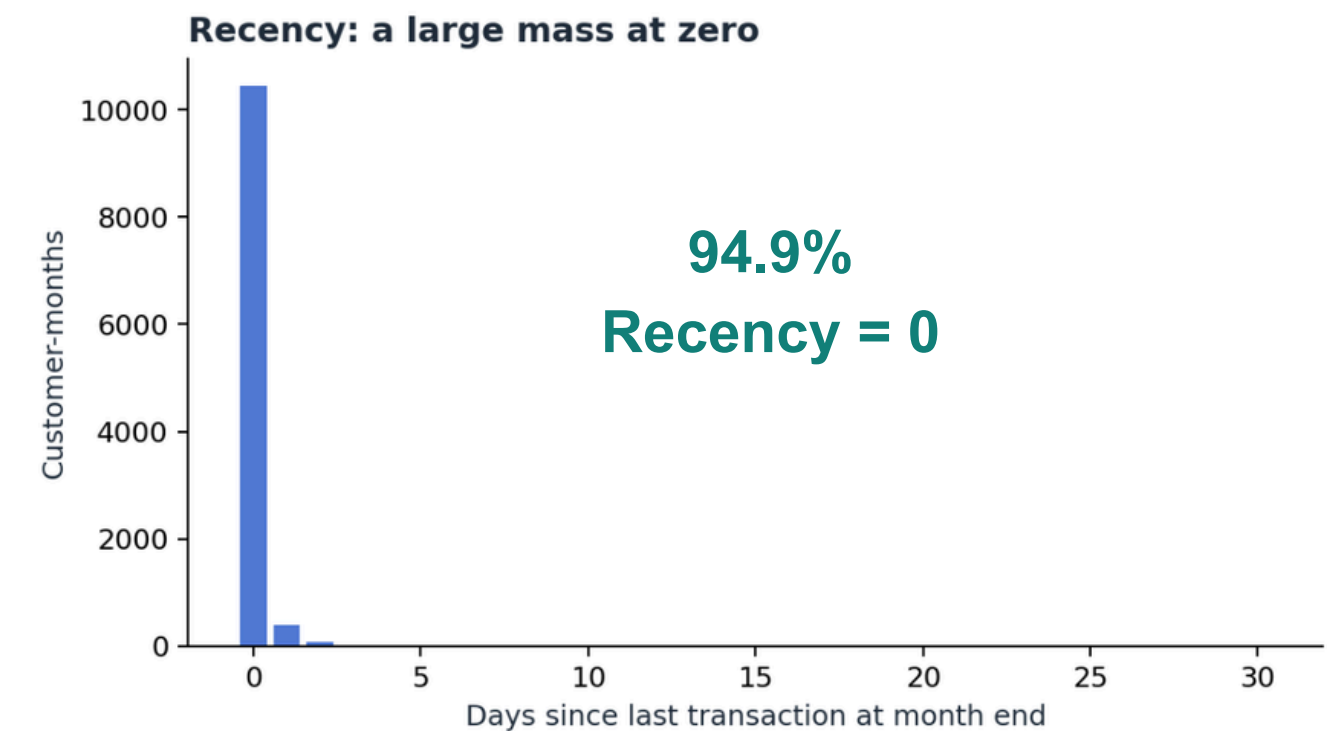
Most customers are already broad and recent - usage intensity is where engagement differs.

All 10,992 observed customer-months: 78.8% use all 14 categories;
frequency still spans 2–713 transactions/month



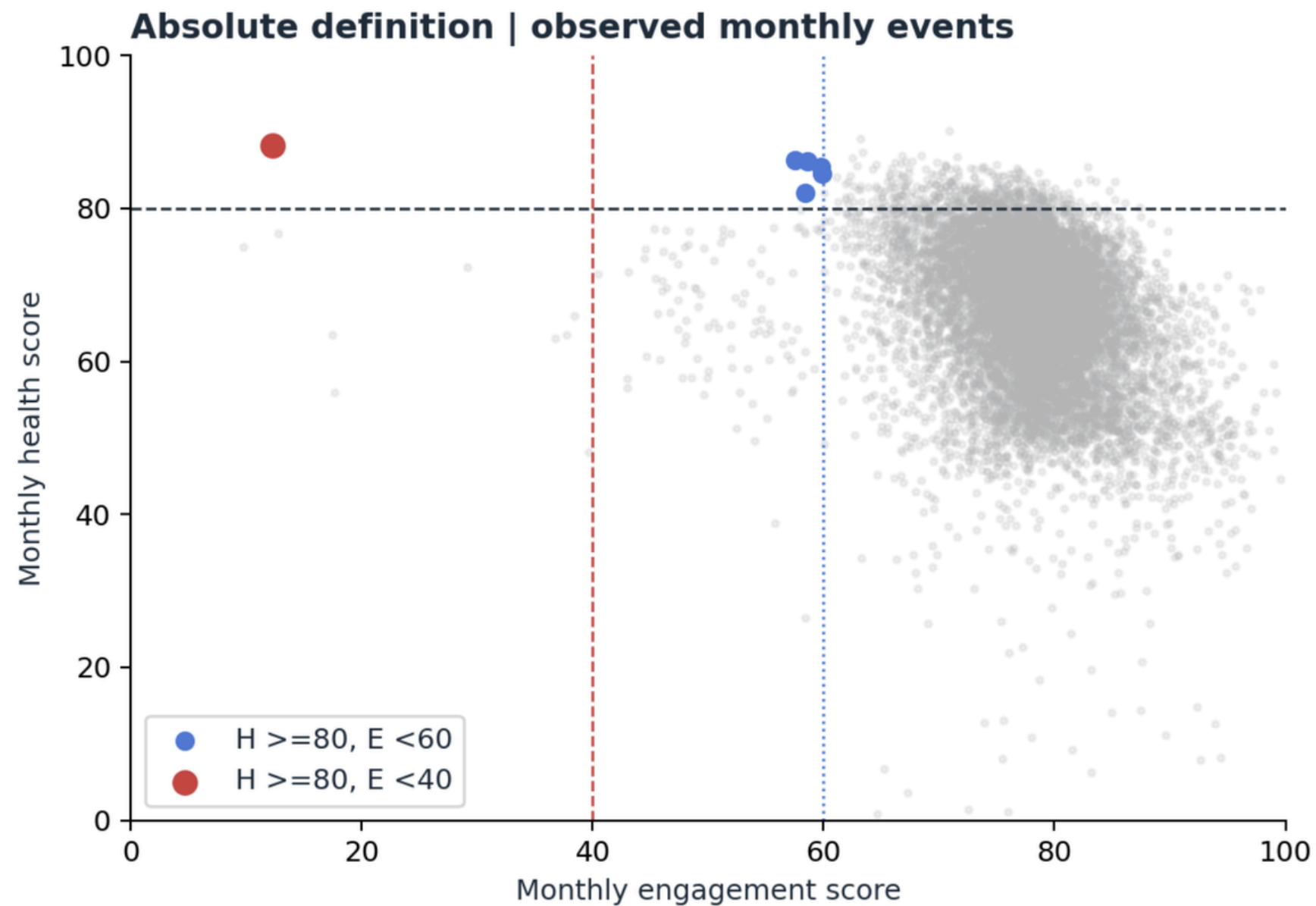
Diversity: 214 categories | Recency: 30 days | Active days: 131 (median 30)

Breadth and recency explain little variation for the majority because both are saturated. Frequency provides clearer behavioral separation, while recency becomes informative mainly in the small inactive tail.



The required healthy - low engagement crossover is almost absent under absolute thresholds.

Absolute monthly screen: health ≥ 80 and engagement < 40 . Relative annual screen: top/bottom quartiles of 908 complete customers.



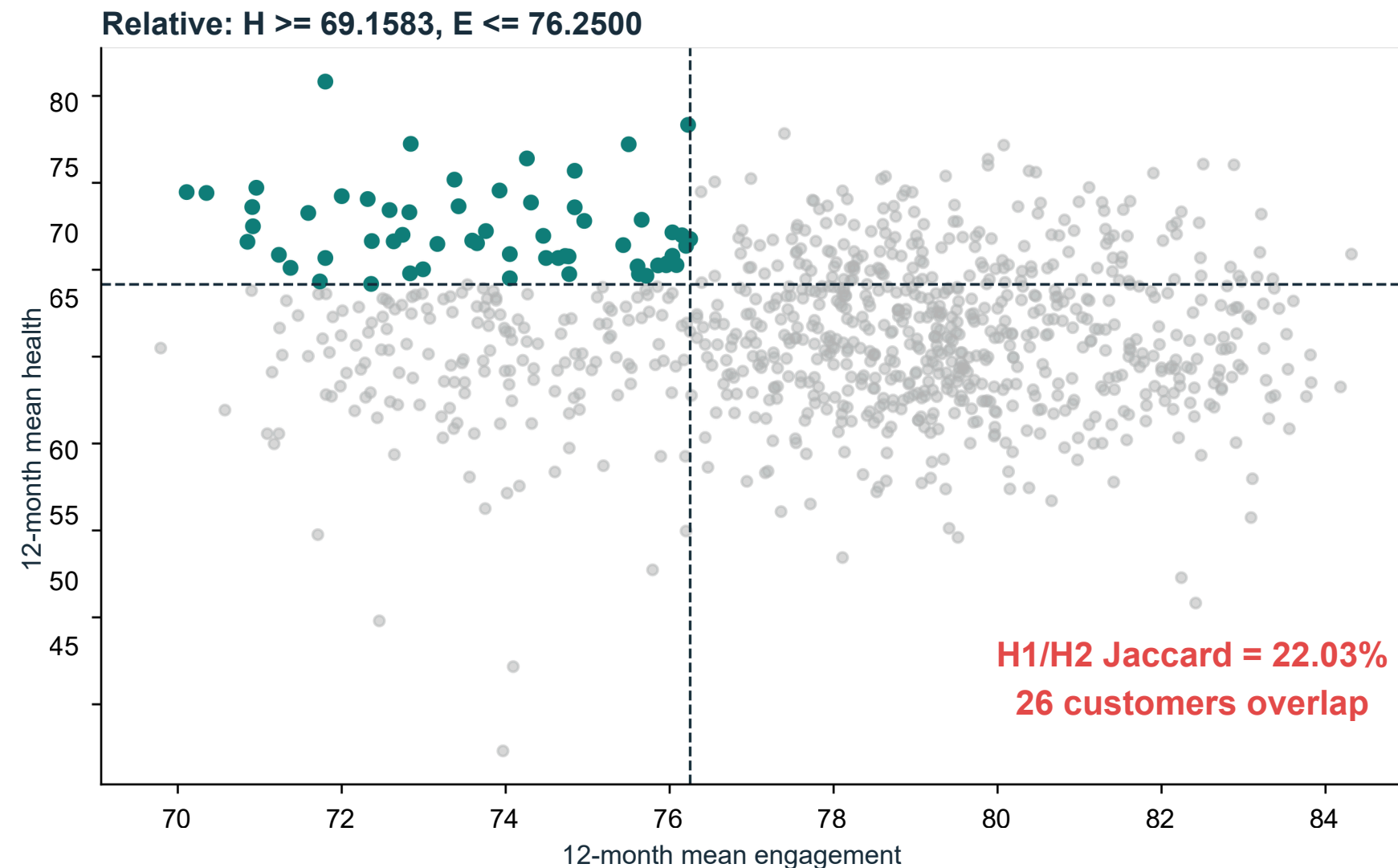
1 / 999 absolute monthly signal

3 transactions; 1 active day; recency 27 days.
E <60 expands to 6 customers,
Including moderate engagement.
No absolute $H \geq 80 / E < 40$ case on full-year averages.

The absence of a large segment is itself a result — thresholds should not be loosened simply to manufacture one.

A relative 6.5% cohort reveals a lower-usage opportunity

But it is dynamic, not a permanent segment.



59 / 908 = 6.50% relative nurture cohort

Frequency

61.6 vs 182.7

Active days

25.4 vs 30.1

Online share

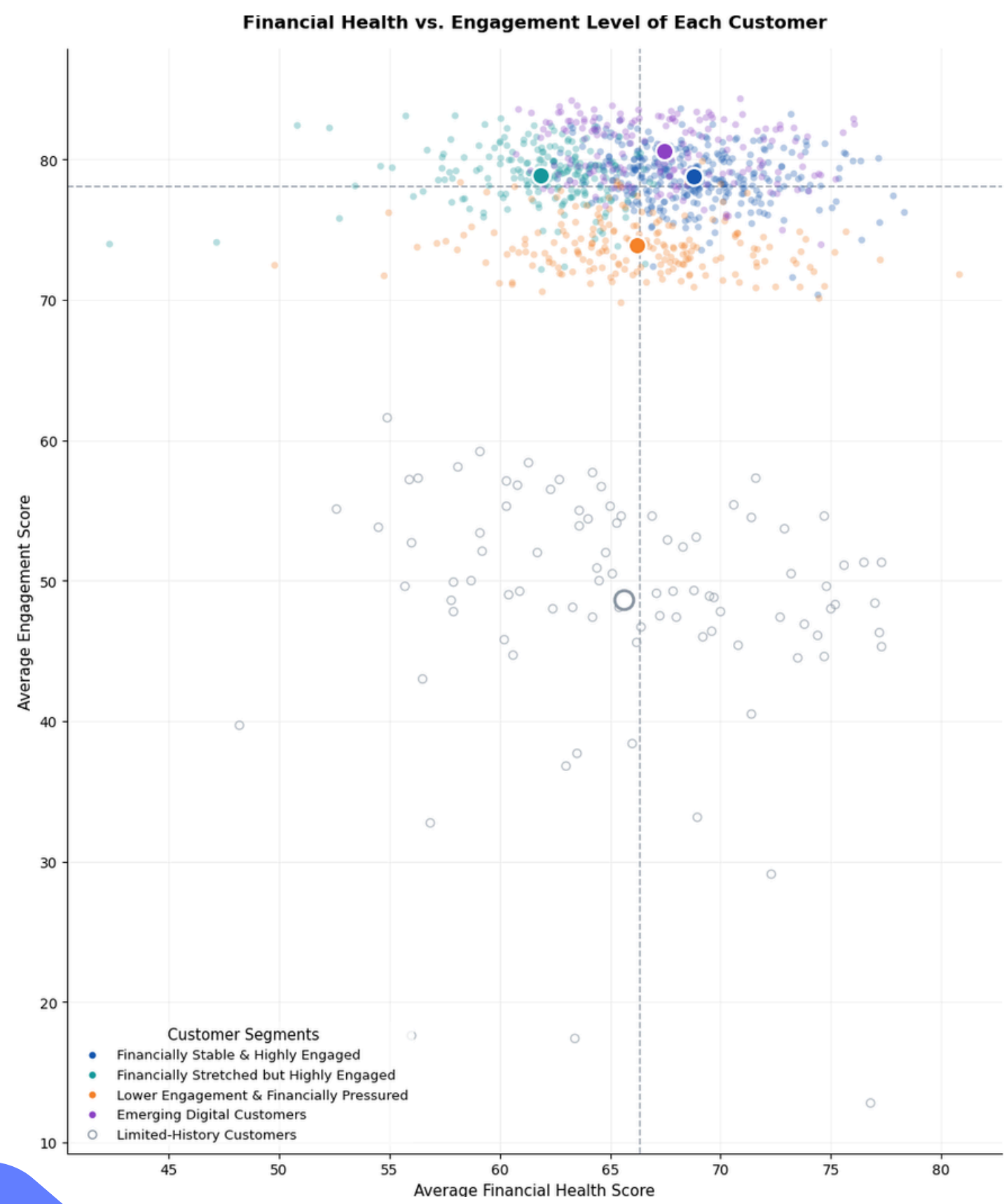
18.55% vs 20.12%

Their engagement gap is primarily lower usage intensity, not lack of channel access.

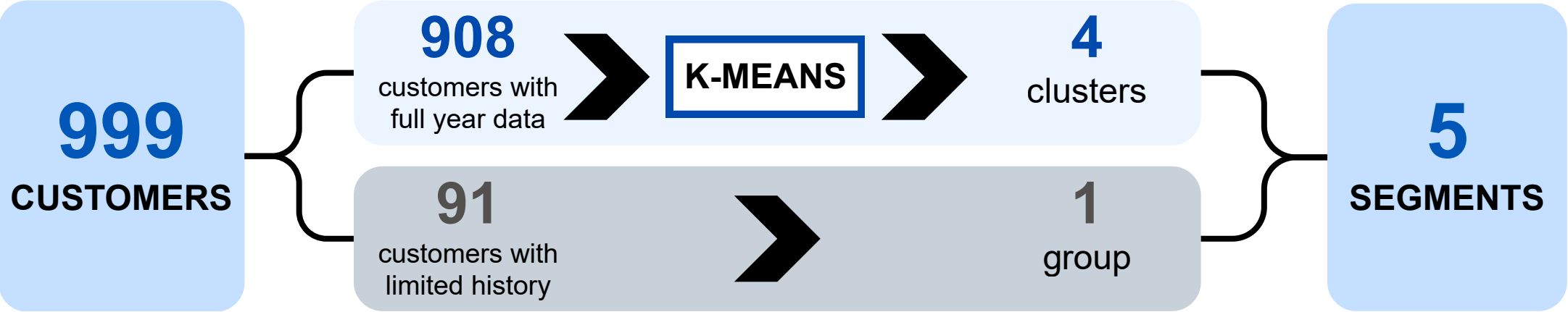
Treat this as a dynamic watchlist, not a permanent customer segment.

From adoption to depth: ITB's opportunity is to increase useful repeat engagement, not simply first-time digital use.

5 SEGMENTS COVER ALL 999 CUSTOMERS THROUGH A HYBRID MODEL



CUSTOMER SEGMENTATION



k = 4: best balance of validation & interpretability (Silhouette = 0.19)

8
CLUSTERING
FEATURES

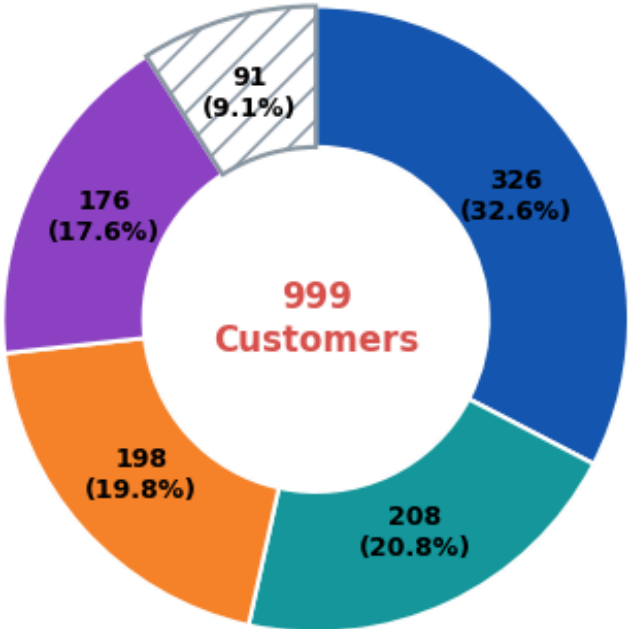
Financial Health

- Spend-to-income ratio
- Credit utilization ratio
- Spending volatility
- Essential spend ratio

Engagement

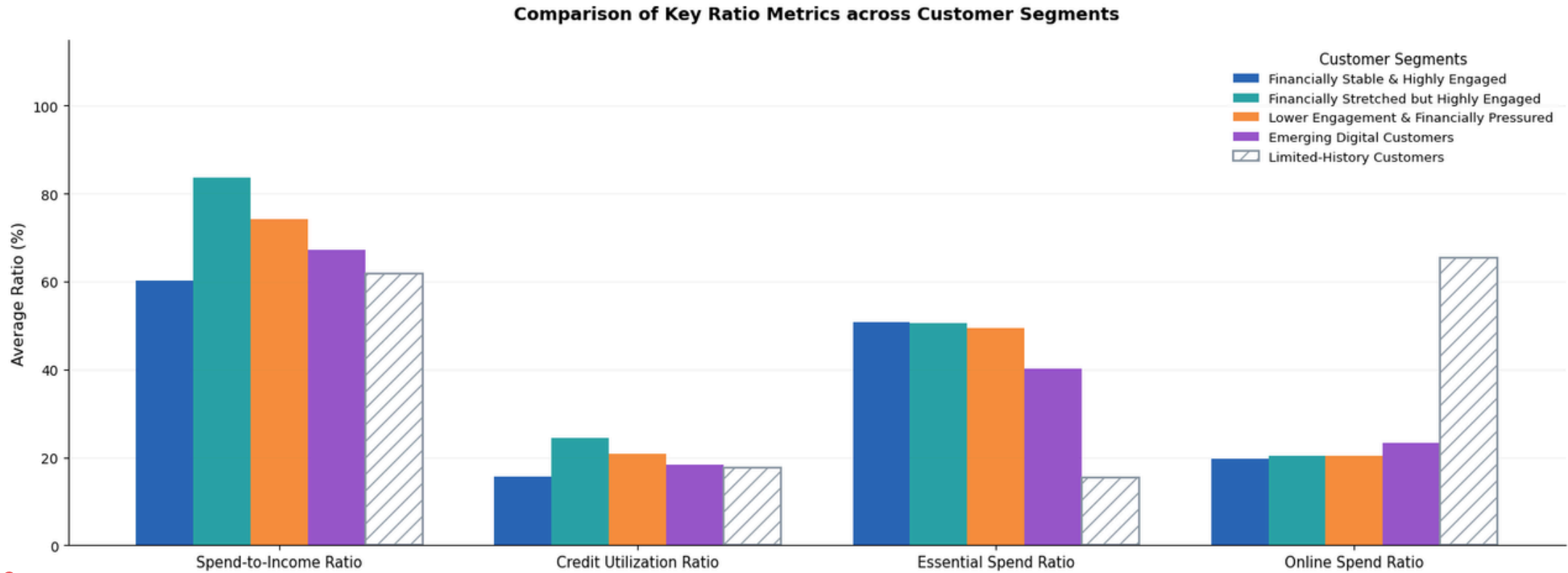
- Transaction count
- Transaction recency
- Online spend ratio
- Category diversity

- Financially Stable & Highly Engaged
- Financially Stretched but Highly Engaged
- Lower Engagement & Financially Pressured
- Emerging Digital Customers
- Limited-History Customers



Large circles indicate segment centers; dashed lines show overall customer means.

FINANCIALLY STRETCHED BUT HIGHLY ENGAGED CUSTOMERS SHOW THE HIGHEST FINANCIAL PRESSURE



Limited-History profiles are descriptive and not directly comparable to full-year customers.

Recommendation Actions by Segment

Financially Stable & Highly Engaged	Build loyalty with spending insights and relevant offers
Financially Stretched but Highly Engaged	Support budgeting with alerts and planning reminders
Lower Engagement & Financially Pressured	Re-engage gently with budgeting support
Emerging Digital Customers	Deepen digital engagement and personalization
Limited-History Customers	Strengthen onboarding and collect more behavioral history

Key Takeaways

Financially Stable & Highly Engaged

balanced spending with **15.5% credit utilization**.

Financially Stretched but Highly Engaged

highest financial pressure, with **83.7% spend-to-income**.

Lower Engagement & Financially Pressured

relatively high spending pressure with weaker digital activity.

Emerging Digital Customers

strongest online preference (**23.2% online spend**).

Limited-History Customers

unusual digital mix (**65.4% online spend**); interpret cautiously.

Limitations

- Moderate cluster separation
- Sensitivity to K-Means assumptions and feature selection
- Loss of temporal detail after annual aggregation
- Limited-History segment is rule-based, not learned by K-Means
- Validation relies mainly on internal clustering metrics

Next steps

- Benchmark alternative clustering methods
- Run feature robustness and weighting tests
- Add temporal / longitudinal features
- Reclassify customers once sufficient history is available
- Validate segments with business outcomes and A/B tests

BUSINESS RECOMMENDATIONS

Six non-punitive actions, each tied to a data finding and an exact target group

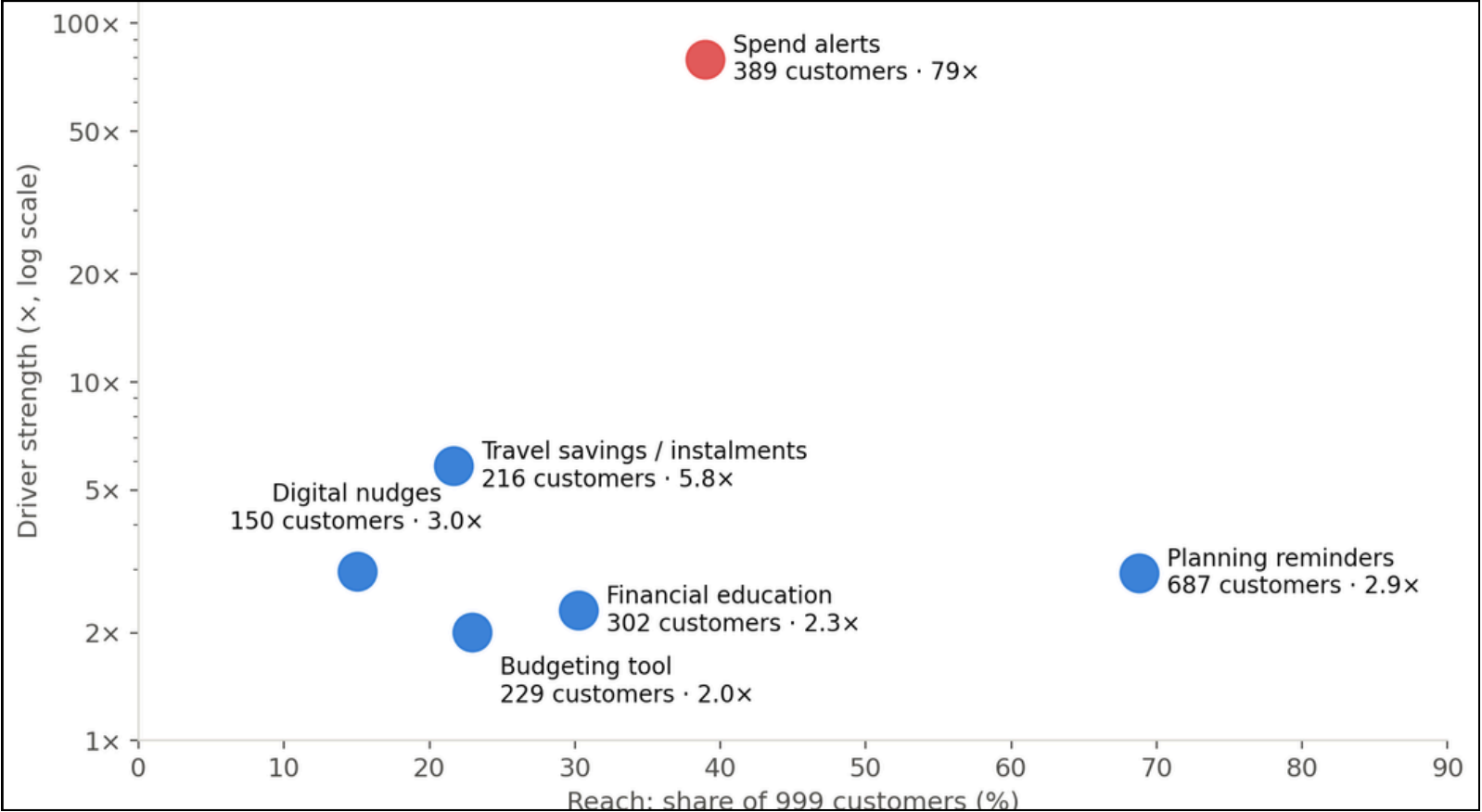
Tool	Finding	Action	Target	Driver
Spend alerts	73.4% of months scoring below 40 contain one purchase $\geq$ 20% of monthly income, vs 0.9% of months scoring 60–80 (spend_amount_vnd, monthly_income_vnd)	Real-time alert at checkout when one purchase reaches 20% of the customer's own monthly income	389 (38.9%) made at least one purchase $\geq$ 20% of monthly income; start with the 61 of them who also had a month below 40	79×
Planning reminders	December and January hold 48.9% of months scoring below 40, 2.9× the 16.7% expected if every month were equally likely (analysis_month, financial_health_score)	Reminder from November to plan December and Tet spending	687 (68.8%): spent more than their income in December (spend_to_income_ratio)	2.9×
Budgeting tools	Customers in the Stretched & Engaged segment or the stressed-but-engaged group spend more than they earn in 2.59 months a year, vs 1.29 for other full-year customers (spend_to_income_ratio > 1)	In-app monthly budget with an alert at 80% of income	229 customers (22.9%) who are in the Stretched & Engaged segment, the stressed-but-engaged group, or both	2.0×
Financial education	12.6% of customers in media & arts, public sector and education have had a month below 40, vs 5.5% in other fields (occupation, financial_health_score)	Three short in-app lessons: keeping a savings buffer, planning purchases above 20% of monthly income, and balancing essential vs discretionary spending	302 (30.2%): all customers in these three fields	2.3×
Digital-channel nudges	Healthy but less engaged customers make 61.6 transactions a month, vs 182.7 for other full-year customers (transaction_count, transaction_channel)	Help set up QR and recurring bills in app; welcome journey for customers with limited history	150 (15.0%): the 59 healthy-but-less-engaged and the 91 with only 1–2 months of data	3.0×
Product suggestions	Travel is 23.7% of spending in months scoring below 40, vs 4.1% in months scoring 60–80 (spending_category, spend_amount_vnd)	Travel is 23.7% of spending in months scoring below 40, vs 4.1% in months scoring 60–80 (spending_category, spend_amount_vnd)	216 (21.6%) made a travel purchase $\geq$ 20% of monthly income; 213 aged 18+ can see instalments	5.8×

Definition

Driver = how many times more often the problem shows up in the target group than in other customers (73.4% ÷ 0.9% = 79×). Target sizes count all 999 customers; findings use the 908 with a full year of data. The budgeting and digital-nudge drivers are less reliable, because those groups were partly chosen using the same measure.

PRIORITY AND THE CREDIT RULE

Start with spend alerts:
the strongest driver and the second-widest reach



Rollout order

Spend alerts

strongest driver (79x), 389 customers

Planning reminders

widest reach (687), one seasonal send from November

Budgeting tool

catches steady month-by-month overspending that alerts and reminders miss

Travel savings / instalments

second-strongest driver (5.8x); opt-in, 18+ only

Financial education

302 customers, low cost

Digital nudges

smallest group; pilot with a control group first

89.4% of 999 customers receive at least one action. 125 (12.5%) qualify for four or more, so messages follow the order above with a monthly cap

By reach	Planning reminders 687 › Spend alerts 389 › Financial education 302 › Budgeting tool 229 › Travel savings 216 › Digital nudges 150
By driver	Spend alerts 79x › Travel savings 5.8x › Digital nudges 3.0x › Planning reminders 2.9x › Financial education 2.3x › Budgeting tool 2.0x
Credit rule	financial_health_score is used only to decide who gets support first. It is never used to deny credit, cut a credit limit, or block an account.